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Rotating joint

Abstract

A rotary joint capable of measuring a rotation angle with high accuracy includes: a flange; a rotor whose one end is connected with the flange and has a tubular shape; a shaft whose outer periphery is connected with an inner periphery of the rotor so that the shaft is rotatable about a central axis of the rotor as a rotation axis; and a rod whose one end is connected with the flange and is provided in an opening to extend from one end side of the shaft along the central axis; an angle measurement unit connected with the opening to extend from another end side of the shaft to another end of the rod and measures a rotation angle between the rod and the shaft with the central axis as the rotation axis; and a shield member formed in a cup shape.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a rotary joint.

BACKGROUND ART

(2) A rotary joint that conveys liquid flowing through a fixed duct line to a rotating object has been conventionally known.

(3) Regarding the rotary joint, Patent Literature 1 discloses a swivel joint in which a rotation angle of a rod that is secured on one of an inner body and an outer body and that is inserted through a pilot oil passage provided in the inner body is detected by a sensor secured on the other of the inner body and the outer body.

CITATION LIST
Patent Literature
(4) Patent Literature 1: JP 6592327B2
SUMMARY OF INVENTION
Technical Problem
(5) However, according to the technique described in Patent Literature 1, since the sensor faces the oil passage, there is a possibility that adherence of a foreign substance flowing through the passage, a foreign substance contained in oil, or the like to the sensor decreases accuracy of measurement of the rotation angle.
(6) The present invention has been made in consideration of such an issue, and is directed to provision of a rotary joint capable of measuring a rotation angle with high accuracy.
Solution to Problem
(7) To solve the above-mentioned issue, a rotary joint according to the present invention includes: a flange; a rotor whose one end is connected with the flange and that has a tubular shape; a shaft whose outer periphery is connected with an inner periphery of the rotor so that the shaft is rotatable about a central axis of the rotor as a rotation axis, the shaft including an opening formed along the central axis; a rod whose one end is connected with the flange and that is provided in the opening so as to extend from one end side of the shaft along the central axis; an angle measurement unit that is connected with the opening so as to extend from another end side of the shaft to another end of the rod and that measures a rotation angle between the rod and the shaft with the central axis serving as the rotation axis; and a shield member that is formed in a cup shape, an outer periphery of the shield member being connected with the opening so that the shield member is rotatable along the central axis, the other end of the rod being connected with an inner periphery of the shield member, and the shield member acting as a shield between the rod and the angle measurement unit in the opening.
(8) Additionally, the shield member is fitted to an inner periphery of the opening with annular members provided on the outer periphery of the shield member so that the shield member is rotatable along the central axis.
(9) Additionally, the rotary joint according to the present invention further includes a magnet provided between the other end of the rod and an inner bottom surface of the shield member, in which the angle measurement unit measures a change in magnetic field generated from the magnet with a magnetic sensor provided on a surface facing an outer bottom surface of the shield member and measures the rotation angle based on a result of the measurement.
(10) Additionally, the hole serves as a drain oil passage.
(11) Additionally, the flange includes a duct line extending from an outside to the one end of the shaft. The shaft includes a discharge opening that extends from the opening to the outer periphery of the shaft, and one end side of the shaft is connected with the duct line so as that the shaft is rotatable about the central axis as the rotation axis.
Advantageous Effects of Invention
(12) According to the present invention, the rotary joint is capable of measuring the rotation angle with high accuracy.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a view illustrating a construction machine on which a rotary joint according to the present embodiment is mounted.
(2) FIG. **2** is a perspective view illustrating a configuration of the rotary joint illustrated in FIG. **1**.
(3) FIG. **3**A is a side view of the rotary joint illustrated in FIG. **1**.

(4) FIG. **3**B is a top view of the rotary joint illustrated in FIG. **1**.
(5) FIG. **3**C is a bottom view of the rotary joint illustrated in FIG. **1**.
(6) FIG. **4** is a perspective view illustrating a configuration of a connection portion between a flange of the rotary joint and a rod of the rotary joint illustrated in FIG. **1**.
(7) FIG. **5** is a cross-sectional view of the rotary joint illustrated in FIG. **1** along an I-I line.
(8) FIG. **6** is a cross-sectional view of the rod, a magnet, a shield member, and an angle measurement unit, which are illustrated in FIG. **5**, along the I-I line.

DESCRIPTION OF EMBODIMENTS

(9) An embodiment of the present invention (hereinafter referred to as “the present embodiment”) will be described below with reference to the accompanying drawings. In each drawing, an identical component and an identical step are each denoted by an identical reference number as much as possible for facilitating understanding, and an overlapping description is omitted.
(10) FIG. **1** is a view illustrating a construction machine **1** on which a rotary joint **10** according to the present embodiment is mounted. As illustrated in FIG. **1**, the construction machine **1** is, for example, a shovel-loader, a crane truck, or the like, and includes a traveling body **2**, a revolving body **3**, an arm portion **4**, and the rotary joint **10**.
(11) The traveling body **2** is provided, for example, in a lower portion of the construction machine **1**. The traveling body **2** supports the whole of the construction machine **1**, and also causes the construction machine **1** to travel forward and backward according to a control signal, a control instruction, a control operation, or the like transmitted from a cab in the revolving body **3** by an operation by a driver of the construction machine **1**. The traveling body **2** is configured to include, for example, a crawler track such as a caterpillar track, and a hydraulic rotation actuator for rotating a wheel of the crawler track (hereinafter referred to as a hydraulic motor). Additionally, the traveling body **2** is provided with a rotary strut in an upper portion, and the revolving body **3** is connected with the traveling body **2** so that the revolving body **3** is rotatable about an axis perpendicular to an installation surface of the traveling body **2** as a rotation axis via the rotary strut. Additionally, the rotary joint **10**, which is formed so that an upper portion of the rotary joint **10** and a lower portion thereof are mutually rotatable about a central axis as a rotation axis, is connected with the upper portion of the traveling body **2**. The traveling body **2** is supplied with oil for hydraulic control from the revolving body **3** via the rotary joint **10**.
(12) The revolving body **3** is provided above the traveling body **2**, and is connected with the traveling body **2** so that the revolving body **3** is rotatable with respect to the traveling body **2** about the axis perpendicular to the installation surface of the traveling body **2** as the rotation axis via the rotary strut connected with the lower portion of the revolving body **3**. The revolving body **3** is configured to include, for example, the cab on which the driver of the construction machine **1** rides and that is used for controlling an operation of the construction machine **1**, and a hydraulic pump for supplying oil to each component that operates with hydraulic pressure of the construction machine **1**. The arm portion **4** is connected with the upper portion of the revolving body **3**. The revolving body **3** supplies oil to the arm portion **4** from the hydraulic pump. Additionally, the rotary joint **10** is connected with the lower portion of the revolving body **3**. The revolving body **3** supplies oil to the traveling body **2** from the hydraulic pump via the rotary joint **10**.
(13) The arm portion **4** is, for example, a multi-joint hydraulic arm provided with a working portion such as a bucket used for construction work at a leading end of the arm portion **4**, and is provided above the revolving body **3**. The arm portion **4** causes each joint portion and the working portion to operate with oil supplied from the hydraulic pump according to the control signal, the control instruction, the control operation, or the like transmitted from the cab of the revolving body **3**.
(14) The rotary joint **10** is, for example, a rotary swivel joint, and supplies oil from the revolving body **3** to the traveling body **2**. The rotary joint **10** is provided between the traveling body **2** and the revolving body **3**. The upper end of the rotary joint **10** is connected with the lower portion of the revolving body **3**. The lower end of the rotary joint **10** is connected with the upper portion of the

traveling body **2**. Additionally, the rotary joint **10** has a tubular shape and is formed so that the upper portion of the rotary joint **10** and the lower portion thereof are mutually rotatable about the central axis of the tube as the rotation axis. Furthermore, the rotary joint **10**, to supply oil to the traveling body **2**, discharges drain oil in the hydraulic motor provided in the traveling body **2** to a tank provided in the revolving body **3**.

(15) Subsequently, a configuration of the rotary joint **10** is described. FIG. **2** is a perspective view illustrating a configuration of the rotary joint **10** illustrated in FIG. **1**. Additionally, FIG. **3**A is a side view of the rotary joint **10** illustrated in FIG. **1**. FIG. **3**B is a top view of the rotary joint **10** illustrated in FIG. **1**. Furthermore, FIG. **4** is a perspective view illustrating a configuration of a connection portion between a flange **11** of the rotary joint **10** and a rod **14** of the rotary joint **10** illustrated in FIG. **1**. Moreover, FIG. **5** is a cross-sectional view of the rotary joint **10** illustrated in FIG. **1** along an I-I line.

(16) As illustrated in FIGS. **2**, **3**A to **3**C, **4**, and **5**, the rotary joint **10** is configured to include, for example, the flange **11**, a rotor **12**, a shaft **13**, the rod **14**, a magnet **15**, a shield member **16**, an angle measurement unit **17**, a fixing member **18**, and a pin **19**. Note that in the present embodiment, assume that a direction that is perpendicular to an installation surface of the rotary joint **10** and that extends from a lower end side of the rotary joint **10** to an upper end side of the rotary joint **10** along a central axis O is a Z-axis direction. Additionally, assume that a direction that is perpendicular to a Z-axis and that extends from a central portion of the flange **11** to one of two ends of a plate-like portion of the flange **11** where run-through holes **113**A and **113**B exist is a Y-axis direction. Furthermore, assume that a direction that is perpendicular to the Z-axis and a Y-axis and that extends from a connection port **116**A on an outer periphery portion of a duct line **111**B in the flange **11** to a connection port on an outer periphery portion of a duct line **111**A is an X-axis direction.

(17) A structure of the FIG. **11** is now described with reference to FIGS. **4** and **5**. The flange **11** is a member for fixing the rotary joint **10** to the installation surface of the rotary joint **10**. The flange **11** is configured to include, for example, the plate-like portion and a tubular portion that is formed so that one of two ends of the tubular portion extends in the X-axis direction and the other thereof extends in an opposite direction of the X-axis direction. Additionally, the flange **11** is provided with a circular recess portion **112** in the central portion of the plate-like portion when viewed from the Z-axis direction. Furthermore, the flange **11** is provided with an opening **115** that penetrates the plate-like portion and the tubular portion from a top surface of the flange **11** and a bottom surface of the flange **11** at the center of the recess portion **112**. The recess portion **112** functions as part of a drain oil passage for causing drain oil to flow into the rotary joint **10**.

(18) The flange **11** is provided with two connection ports of connection ports **117**A and **117**B around the opening **115** in the recess portion **112**. The flange **11** is provided with the connection port **116**A on a surface at one end on the X-axis direction side out of two ends of the tubular portion. The connection port **116**A leads to the connection port **117**A via the duct line **111**A. Additionally, the flange **11** is provided with a connection port **116**B on a surface at the other end on the opposite side of the X-axis direction, out of the two ends of the tubular portion. The connection port **116**B leads to the connection port **117**B via the duct line **111**B. The duct line **111**A extending from the connection port **116**A to the recess portion **112** via the connection port **117**A functions as the drain oil passage for causing drain oil to flow into the rotary joint **10**. Additionally, the duct line **111**B extending from the connection port **116**B to the recess portion **112** via the connection port **117**B functions as the drain oil passage for causing drain oil to flow into the rotary joint **10**.

(19) One end of the rod **14** is inserted into the opening **115** from the Z-axis direction and is connected with the flange **11** via the fixing member **18**. Specifically, in the flange **11**, the fixing member **18** is fitted to the opening **115** on a bottom surface side, which is the opposite side of the Z-axis direction. The fixing member **18** mentioned herein is a member for fixing and connecting the rod **14** to/with the flange **11**, and is formed to have a cap shape. The fixing member **18** is provided with an annular member **181** on an outer periphery of the fixing member **18**. The annular

member **181** is, for example, an O-ring formed of resin, and is connected with the fixing member **18** so as to encircle the outer periphery of the fixing member **18**. The outer periphery of the fixing member **18** is fitted to an inner periphery of the opening **115** on the bottom surface side via the annular member **181**.
(20) Additionally, an outer periphery of the rod **14** is fitted to an inner periphery of the fixing member **18**. The fixing member **18** is provided with an opening **182** in a fitting portion to which the rod **14** is fitted. The opening **182** penetrates from the inner periphery of the fixing member **18** to the outer periphery of the fixing member **18**. Additionally, an opening **141** having a diameter that is equal to a diameter of the opening **182** provided in the fixing member **18** is provided at one end of the rod **14**. In the fixing member **18**, the pin **19** is fitted to the opening **182** and the opening **141** so as to penetrate the opening **182** and the opening **141**. Furthermore, the fixing member **18** is fitted to or screwed into the opening **115** of the flange **11** from the bottom surface side, and thereby fixes and connects the rod **14** to/with the flange **11**.
(21) The flange **11** is provided with the run-through holes **113**A and **113**B at one end on the Y-axis direction side out of the two ends of the plate-like portion. Additionally, the flange **11** is provided with run-through holes **113**C and **113**D at the other end on the opposite side of the Y-axis direction out of the two ends of the plate-like portion. In the flange **11**, respective bolts are inserted into the run-through holes **113**A to **113**D, and the bolts are inserted into respective mounting holes in a top surface of the traveling body **2**, whereby the bottom surface of the flange **11** is connected with the upper portion of the traveling body **2**. Additionally, the flange **11** is provided with run-through holes **114**A to **114**D so as to surround the holes in the plate-like portion. In the flange **11**, respective bolts are inserted into the run-through holes **114**A to **114**D, and the bolts are inserted into respective female threads at one end of the rotor **12**, whereby one end of the rotor **12** is connected with the top surface of the flange **11**.
(22) A structure of the rotor **12** is now described with reference to FIGS. **3**A, **3**C, and **5**. The rotor **12** is a member formed in a tubular shape, and supplies oil flowing from the shaft **13** to members that are connected with the outer periphery of the rotor **12** and that are provided in the traveling body **2** via the oil passage provided inside the rotor **12**. Examples of oil flowing into the shaft **13** include supply oil to a hydraulic motor, return oil from the hydraulic motor, and pilot oil. Additionally, examples of the oil passage provided inside the rotor **12** include a supply oil passage to the hydraulic motor, a return oil passage from the hydraulic motor, and a pilot oil passage. One end of the rotor **12** on the bottom surface side in the opposite direction of the Z-axis direction out of two ends of the rotor **12** is connected with the top surface side of the plate-like portion of the flange **11**. Additionally, the outer periphery of the shaft **13** is connected with the inner periphery of the rotor **12** so that the rotor **12** is rotatable about the central axis O as the rotation axis.
(23) The rotor **12** is provided with oil passages **121**A to **121**H between the outer periphery and inner periphery of the rotor **12**. The oil passages **121**A to **121**H are used to supply oil flowing from the shaft **13** to the members that are connected with the outer periphery of the rotor **12** and that are provided in the traveling body **2**. The oil passages **121**A to **121**H are duct lines that lead from respective oil passages on the inner periphery of the rotor **12** to respective connection ports on the outer periphery of the rotor **12**. The oil passages **121**A to **121**H are connected with respective connection ports that are associated with the respective oil passages on the inner periphery and that are disposed on the outer periphery of the shaft **13**. Additionally, the respective connection ports on the outer periphery for the oil passages **121**A to **121**H are connected with respective connection ports of members associated with the oil passages **121**A to **121**H out of the members provided in the traveling body **2**.
(24) A structure of the shaft **13** is now described with reference to FIGS. **3**A, **3**B, and **5**. The shaft **13** is a member formed in a tubular shape, and supplies oil flowing from members that are connected with a top surface of the shaft **13** and that are provided in the revolving body **3** to the rotor **12** connected with the outer periphery of the shaft **13** via the oil passages provided inside the

shaft **13**. Additionally, the shaft **13** discharges drain oil flowing from the flange **11** from a discharge opening **134** provided in the outer periphery of the shaft **13** via an opening **132** provided inside the shaft **13**.

(25) The outer periphery of the shaft **13** in a lower portion on the opposite side of the Z-axis direction is connected with the inner periphery of the rotor **12** so that the shaft **13** is rotatable about the central axis O of the rotor **12** as the rotation axis. Additionally, the shaft **13** is provided with a recess portion **133** at one end on the lower portion side out of the two ends of the shaft **13**. The recess portion **133** centers on the central axis O. One end of the shaft **13** on the opposite side of the Z-axis direction is coupled with the top surface side of the flange **11** so that the shaft **13** is rotatable about the central axis O as the rotation axis. Furthermore, in the shaft **13**, the recess portion **133** is connected with the duct lines **111**A and **111**B of the flange **11** so that the shaft **13** is rotatable about the central axis O as the rotation axis. Additionally, in the shaft **13**, formed is the opening **132** extending from the recess portion **133**, along the central axis O, to the other end of the shaft **13** in the upper portion on the Z-axis direction side out of the two ends of the shaft **13**. Additionally, the shaft **13** includes the discharge opening **134** that extends from the opening **132** to the outer periphery in the upper portion of the shaft **13**. The opening **132**, the recess portion **133**, and the discharge opening **134**, together with the duct lines **111**A and **111**B in the flange **11**, function as the drain oil passage.

(26) The shaft **13** is provided with connection ports **131**A to **131**D around the opening **132** on the top surface of the shaft **13** in the upper portion. The connection ports **131**A to **131**D are used to circulate oil between the shaft **13** and the revolving body **3**. Additionally, the shaft **13** is provided with connection ports **131**E to **131**H on the outer periphery of the shaft **13** in the upper portion. The connection ports **131**E to **131**H are used to circulate oil between the shaft **13** and the revolving body **3**. The connection ports **131**A to **131**H lead to, via the oil passages provided inside the shaft **13**, respective oil passages associated with the connection ports **131**A to **131**H, out of a plurality of oil passages provided in the outer periphery of the shaft **13** in the lower portion.

(27) In the shaft **13**, the rod **14** is provided in the opening **132** from the one end side of the shaft **13**, which is the opposite side of the Z-axis direction, via the recess portion **112** of the flange **11** and the recess portion **133** of the shaft **13** so that the rod **14** is not in direct contact with the inner periphery of the opening **132**. Additionally, the angle measurement unit **17** is connected with the shaft **13** from the other end side, which is the Z-axis direction side, for example, by being screwed into the opening **132**. Furthermore, in the shaft **13**, the outer periphery of the shield member **16** formed in a cup shape to cover the outer periphery of the rod **14** at the other end is fitted to the inner periphery of the opening **132** so that the shield member **16** is rotatable about the central axis O as the rotation axis.

(28) The rod **14**, the magnet **15**, the shield member **16**, and the angle measurement unit **17** are now described with reference to FIGS. **4** to **6**. The rod **14** is formed in a columnar shape, and one end of the rod **14** on the opposite side of the Z-axis direction is connected with the inner periphery of the fixing member **18**, whereby the rod **14** is connected with the opening **115** of the flange **11** via the fixing member **18** and is provided inside the opening **132** so as to extend from one end side of the shaft **13** along the central axis O. Additionally, the outer periphery of the rod **14** at the other end in the Z-axis direction is connected with an inner periphery **162** of the shield member **16** formed in the cup shape by fitting or screwing. The magnet **15** is provided between a top surface of the rod **14** at the other end and an inner bottom surface **163** of the shield member **16**. Furthermore, the rod **14** is provided in the opening **132** so that a portion between one end of the rod **14** and the other end of the rod **14** is not in direct contact with the inner periphery of the opening **132**.

(29) The magnet **15** is, for example, a permanent magnet, and generates a magnetic field. The magnet **15** is provided between the inner bottom surface **163** of the shield member **16** and the top surface of the rod **14** at the other end. The magnet **15** is used by the angle measurement unit **17** to measure a rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the

rotation axis.
(30) The shield member **16** is made of a non-magnetic material and formed in a cup shape, and acts as a shield between the rod **14** and the angle measurement unit **17** in the opening **132** of the shaft **13**. Additionally, an opening is formed in a central portion of the bottom surface of the shield member **16**. The outer periphery of the rod **14** at the other end is connected with the inner periphery **162** of the shield member **16** by fitting or screwing. Additionally, the magnet **15** is provided between the inner bottom surface **163** of the shield member **16** and the top surface of the rod **14** at the other end so that a surface of the magnet **15** in the Z-axis direction is exposed from an opening provided in the central portion of the bottom surface of the shield member **16**. Note that a sealing material is applied to the inner periphery **162** of the shield member **16** to maintain oil-tightness of the magnet **15**. Additionally, the shield member **16** is provided with annular members **165**A and **165**B on an outer periphery **161** so as to encircle the outer periphery **161**. The shield member **16** is fitted to the inner periphery of the opening **132** via the annular members **165**A and **165**B so that the outer periphery **161** is rotatable along the central axis O. Additionally, an outer bottom surface **164** of the shield member **16** faces a measurement surface of the angle measurement unit **17** in the opening **132**.
(31) Each of the annular members **165**A and **165**B is, for example, an O-ring formed of resin, and is provided in the shield member **16** so as to encircle the outer periphery **161** of the shield member **16**. The annular members **165**A and **165**B are provided between the shield member **16** and the inner periphery of the opening **132** so as to generate such friction as to allow the shaft **13** to rotate with respect to the rod **14** about the central axis O as the rotation axis to such a degree as that drain oil flowing on the rod **14** side in the opening **132** of the shaft **13** does not leak to the one side of the shaft **13** with which the angle measurement unit **17** is connected.
(32) The angle measurement unit **17** is, for example, a magnet-type angle sensor, and measures a rotation angle between the rod **14** and the shaft **13** with the central axis O serving as the rotation axis. The angle measurement unit **17** transmits the measured rotation angle to a control device of the construction machine **1** in the revolving body **3**, the traveling body **2**, or the like. The angle measurement unit **17** includes a magnetic sensor **171**, a socket **172**, and a fixing member **173**.
(33) The socket **172** is formed in a tubular shape, and an outer periphery of the socket **172** is connected with the inner periphery of the opening **132** by, for example, screwing so that the socket **172** extends from the other end side of the shaft **13** to the other end of the rod **14**. A surface of the socket **172** at one end on the opposite side of the Z-axis direction out of two ends of the socket **172** is a measurement surface, which is provided with the magnetic sensor **171**. The measurement surface faces the magnet **15** on the shield member **16** and the outer bottom surface **164** of the shield member **16**, and is located at a distance of, for example, about 3.0 mm from the surface of the magnet **15** on the Z-axis direction side. Additionally, the outer periphery of the fixing member **173** is connected with the inner periphery of the socket **172** at the other end, which is the Z-axis direction side, out of the two ends of the socket **172**. The socket **172** transmits a signal output from the magnetic sensor **171** to the control device of the construction machine **1** in the revolving body **3**, the traveling body **2**, or the like via a signal line that is provided inside the socket **172**, the fixing member **173**, and an external communication cable.
(34) The magnetic sensor **171** is provided on the measurement surface of the angle measurement unit **17**, and measures a change in magnetic field generated from the magnet **15**. The magnetic sensor **171** measures the rotation angle between the rod **14** and the shaft **13** with the central axis O serving as the rotation axis from the measured change in magnetic field. The magnetic sensor **171** uses a result of the measurement as an electric signal, outputs the electric signal to the control device of the construction machine **1** in the revolving body **3**, the traveling body **2**, or the like via the signal line in the socket **172**, the fixing member **173**, and the external communication cable.
(35) The fixing member **173** is a member for fixing the communication cable that connects the rotary joint **10** and the control device of the construction machine **1** to the rotary joint **10**. The outer

periphery of the fixing member **173** at one end is connected with the inner periphery of the socket **172** at the other end on the Z-axis direction side out of the two ends of the socket **172**, by screwing, fitting, or the like. The other end of the fixing member **173** is connected with the communication cable.
(36) <Effects>
(37) As described above, in the present embodiment, the rotary joint **10** includes the flange **11**, the tubular rotor **12** whose one end is connected with the flange **11**, and the shaft **13** whose outer periphery is connected with the inner periphery of the rotor **12** so as to be rotatable about the central axis O of the rotor **12** as the rotation axis and in which the opening **132** is formed along the central axis O. Additionally, the rotary joint **10** includes the rod **14** whose one end is connected with the flange **11** and that is provided in the opening **132** so as to extend from the one end side of the shaft **13** along the central axis O, and the angle measurement unit **17** that is connected with the opening **132** so as to extend from the other end side of the shaft **13** to the other end of the rod **14** and that measures the rotation angle between the rod **14** and the shaft **13** with the central axis O serving as the rotation axis. Furthermore, the rotary joint **10** further includes the shield member **16** that is formed in the cup shape. The outer periphery **161** is connected with the opening **132** so that the shield member **16** is rotatable along the central axis O. The other end of the rod **14** is connected with the inner periphery **162**. The shield member **16** acts as a shield between the rod **14** and the angle measurement unit **17** in the opening **132**.
(38) With this configuration, in the rotary joint **10**, the shield member **16** acts as a shield between the rod **14** side and the angle measurement unit **17** side in the opening **132**. As a result, a foreign substance flowing on the rod **14** side in the opening **132** is prevented from adhering to the angle measurement unit **17**. Hence, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis with high accuracy.
(39) Additionally, the shield member **16** is fitted to the inner periphery of the opening **132** with the annular members **165**A and **165**B provided on the outer periphery **161** of the shield member **16** so that the shield member **16** is rotatable along the central axis O.
(40) With this configuration, in the rotary joint **10**, the annular members **165**A and **165**B adjust the extending direction of the rod **14** to the direction of the central axis O of the shaft **13**. Hence, the rotary joint **10** is capable of preventing the extending direction of the rod **14** from being inclined with respect to the central axis O of the shaft **13**. Additionally, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis using a smaller number of components.
(41) Additionally, the rotary joint **10** further includes the magnet **15** provided between the other end of the rod **14** and the inner bottom surface **163** of the shield member **16**. Additionally, the angle measurement unit **17** measures the change in magnetic field generated from the magnet **15** with the magnetic sensor **171** provided on a surface facing the outer bottom surface **164** of the shield member **16**, and measures the rotation angle based on a result of the measurement.
(42) With this configuration, in the rotary joint **10**, a foreign substance flowing on the rod **14** side in the opening **132** is prevented from adhering to, in addition to the angle measurement unit **17**, the magnet **15**. Hence, the rotary joint **10** measures the rotation angle based on the change in magnetic field, and can thereby measure the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis with high accuracy.
(43) Additionally, the shield member **16** is formed of the non-magnetic material.
(44) Hence, since the shield member **16** is formed of the material that has less influence on a magnetic field, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis with higher accuracy.
(45) Additionally, the opening **132** serves as the drain oil passage.
(46) With this configuration, since the opening **132** serves as the drain oil passage, hydraulic pressure applied to the shield member **16** in the opening **132** decreases and strength necessary for

the shield members **16** and the annular members **165**A and **165**B is suppressed in comparison with a case where the opening **132** serves as, for example, an oil passage other than the drain oil passage such as the pilot oil passage. Hence, since strength necessary for the shield member **16** and the annular members **165**A and **165**B is suppressed, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis at low cost.

(47) Additionally, the flange **11** includes the duct line extending from the outside to one end of the shaft **13**. Furthermore, the shaft **13** includes the discharge opening **134** that extends from the opening **132** to the outer periphery of the shaft **13**, and one end side of the shaft **13** in the opening **132** is connected with the duct line so that the shaft **13** is rotatable about the central axis O as the rotation axis.

(48) That is, in the rotary joint **10**, the rod **14** is provided in the opening **132** out of the passage extending from the outside of the flange **11** to the discharge opening **134** via the opening **132**. With this configuration, in the rotary joint **10**, the opening **132** provided with the rod **14** is used as the passage, whereby it is possible to suppress an increase in size of the rotary joint **10**.

(49) <Modifications>

(50) Note that the present invention is not limited to the above-described embodiment. That is, a modification obtained by addition of a design change to the above-mentioned embodiment as appropriate by the person skilled in the art is included in the scope of the present invention as long as the modification has the features of the present invention. Additionally, each element included in the above-mentioned embodiment and the modification, which will be described later, may be combined as long as it is technically possible, and a combination thereof is included in the scope of the present invention as long as the combination has the features of the present invention.

(51) For example, in the present embodiment, the rod **14** is formed in the columnar shape, but the shape of the rod **14** is not limited thereto. The rod **14** may have any shape as long as, for example, the rod **14** extends from the flange **11** to the shield member **16** and the leading end of the rod **14** is connected with the inner periphery **162** of the shield member **16**. The rod **14** may be formed, for example, in a prism shape. In a case where the rod **14** is formed in the prism shape, the inside of the shield member **16** is formed in a shape according to a shape of the leading end of the rod **14**.

(52) With this configuration, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis with high accuracy regardless of the shape of the rod **14**.

(53) Additionally, the shield member **16** is provided with the annular members **165**A and **165**B, such as O-rings, on the outer periphery **161**, but the configuration is not limited thereto. An oil seal, a cap seal, or the like that is formed so that the surroundings of the annular members formed of metal are covered with resin may be provided on the outer periphery **161** of the shield member **16**.

(54) With this configuration, since a member other than the O-ring is used as a member for sealing oil in the shield member **16**, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis with high accuracy also in a case where the member other than the O-ring such as the oil seal and the cap seal is appropriate in terms of cost or the like.

(55) Additionally, the shield member **16** may be subjected to coating treatment with plating or the like. Additionally, a portion of the rod **14** in contact with the inner periphery **162** of the shield member **16** may be subjected to coating treatment with plating or the like.

(56) According to the configuration, since abrasion of the rod **14** and the shield member **16** is reduced by the coating treatment, the rotary joint **10** is capable of measuring the rotation angle between the shaft **13** and the rod **14** with the central axis O serving as the rotation axis with high durability and high accuracy.

REFERENCE SIGNS LIST

(57) **10**: Rotary joint **11**: Flange **12**: Rotor **13**: Shaft **14**: Rod **15**: Magnet **16**: Shield member **17**:

Angle measurement unit **111**A: Duct line **111**B: Duct line **131**: Opening **132**: Discharge opening **161**: Outer periphery **162**: Inner periphery **163**: Inner bottom surface **164**: Outer bottom surface **165**A: Annular member **165**B: Annular member **171**: Magnetic sensor

Claims

1. A rotary joint comprising: a flange; a rotor whose one end is connected with the flange and that has a tubular shape; a shaft whose outer periphery is connected with an inner periphery of the rotor so that the shaft is rotatable about a central axis of the rotor as a rotation axis, the shaft including an opening which serves as a drain oil passage formed along the central axis and a discharge opening that extends from the opening to the outer periphery of the shaft; a rod whose one end is connected with the flange and that is provided in the opening so as to extend from one end side of the shaft along the central axis; an angle measurement unit that is connected with the opening so as to extend from another end side of the shaft to another end of the rod and that measures a rotation angle between the rod and the shaft with the central axis serving as the rotation axis; and a shield member that is formed in a cup shape, an outer periphery of the shield member being connected with the opening so that the shield member is rotatable along the central axis, the other end of the rod being connected with an inner periphery of the shield member, and the shield member acting as a shield between the rod and the angle measurement unit in the opening and being provided away from the angle measurement unit.
2. The rotary joint according to claim 1, wherein the shield member is fitted to an inner periphery of the opening with annular members provided on the outer periphery of the shield member so that the shield member is rotatable along the central axis.
3. The rotary joint according to claim 2, further comprising a magnet provided between the other end of the rod and an inner bottom surface of the shield member, wherein the angle measurement unit measures a change in magnetic field generated from the magnet with a magnetic sensor provided on a surface facing an outer bottom surface of the shield member and measures the rotation angle based on a result of the measurement.
4. The rotary joint according to claim 3, wherein the flange includes a duct line extending from an outside to the one end of the shaft, and one end side of the shaft is connected with the duct line so as that the shaft is rotatable about the central axis as the rotation axis.
5. The rotary joint according to claim 1, further comprising a magnet provided between the other end of the rod and an inner bottom surface of the shield member, wherein the angle measurement unit measures a change in magnetic field generated from the magnet with a magnetic sensor provided on a surface facing an outer bottom surface of the shield member and measures the rotation angle based on a result of the measurement.
6. The rotary joint according to claim 5, wherein the flange includes a duct line extending from an outside to the one end of the shaft, and one end side of the shaft is connected with the duct line so as that the shaft is rotatable about the central axis as the rotation axis.
7. A rotary joint comprising: a flange; a rotor whose one end is connected with the flange and that has a tubular shape; a shaft whose outer periphery is connected with an inner periphery of the rotor so that the shaft is rotatable about a central axis of the rotor as a rotation axis, the shaft including an opening which serves as a drain oil passage formed along the central axis; a rod whose one end is connected with the flange and that is provided in the opening so as to extend from one end side of the shaft along the central axis; an angle measurement unit that is connected with the opening so as to extend from another end side of the shaft to another end of the rod and that measures a rotation angle between the rod and the shaft with the central axis serving as the rotation axis; and a shield member that is formed in a cup shape, an outer periphery of the shield member being connected with the opening so that the shield member is rotatable along the central axis, the other end of the rod being connected with an inner periphery of the shield member, and the shield member acting as

a shield between the rod and the angle measurement unit in the opening; wherein the flange includes a duct line extending from an outside to the one end of the shaft, and the shaft includes a discharge opening that extends from the opening to the outer periphery of the shaft, and one end side of the shaft is connected with the duct line so as that the shaft is rotatable about the central axis as the rotation axis.